Result 1 (short pullback identity map). [Topology foundations, inferred carrier comparison]
Let K be a type. Let *field* be a field structure on K. Let V, E, W and F be types. Let Q be a directed shape with vertices in V and edges in E. Let R be a directed shape with vertices in W and edges in F. Let f be a map of directed shapes from Q to R. Let M be a module diagram on R. Let $N = f^*(M)$. Let v be an element of V. Then

$$\mathrm{id}_{N.\mathrm{object}(v)} = \mathrm{id}_{M.\mathrm{object}(f.\mathrm{vertices}(v))}.$$

Result 2 (real constant application). [Topology foundations, inferred binder domain]
Let x and y be real numbers. Then

$$\mathrm{const}(x)(y) = x.$$

Result 3 (function constant application). [Topology foundations, inferred compound binder domain]
Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$. Then

$$\mathrm{const}(f)(g) = f.$$

Result 4 (dependent constant application). [Topology foundations, inferred dependent binder domain]
Let f be a function assigning to each $n : \mathbb{N}$ a value of type $\mathbb{R}^n$. Let g be a function assigning to each $m : \mathbb{N}$ a value of type $\mathbb{R}^m$. Then

$$\mathrm{const}(f)(g) = f.$$

Result 5 (inferred dimension entry). [Topology foundations, inferred dimension comparison]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{R}$. Let B be a matrix of size $\mathrm{size}(A) \times \mathrm{size}(A)$ over $\mathbb{R}$. Assume $B = A$ (`same`). Let i and j be vertices in $\{0, \ldots, n-1\}$. Then

$$B_{i,j} = A_{i,j}.$$

Result 6 (literal matrix size). [Topology foundations, inferred natural literal]
Let A be a matrix of size 3×3 over $\mathbb{R}$. Then

$$\mathrm{size}(A) = 3.$$

LLM-assisted specimens and proofs (gpt-6, 2026-11-06). See PROVENANCE.md. Comparison preserves the authored Lean calls.